

MITOCHONDRIAL DNA SEQUENCE EVIDENCE SUPPORTING THE RECOGNITION OF TWO SUBSPECIES OR SPECIES OF THE FLORIDA SPINY LOBSTER *PANULIRUS ARGUS*

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ABSTRACT

Samples of the spiny lobster *Panulirus argus* from populations in the western Atlantic and Caribbean were compared with spiny lobsters from Brazil for DNA sequence variation for 2 mitochondrial genes. Phylogenetic analyses of 16S sequence data reveal 2 distinct clades of *P. argus*, a clade consisting of Caribbean populations and a second clade of samples from Brazil. Mean sequence divergence estimates between Brazilian and western Atlantic/Caribbean populations were 8.32% for the mitochondrial large subunit ribosomal RNA (16S) and 19.82% for the cytochrome oxidase I subunit (COI). Levels of sequence divergence among sister species of *Panulirus* in the japonicus-group were similar to that observed between Brazilian and Caribbean/western Atlantic *P. argus*. In addition to large mitochondrial DNA sequence differences between western Atlantic/Caribbean and Brazilian *P. argus*, characteristic color patterns also distinguish Brazilian *P. argus*. Based on the results presented here, Brazilian *P. argus* is distinct from populations in the Caribbean and western Atlantic and may represent a complex of 2 species or subspecies. The degree of differentiation between the Caribbean and Brazilian clades indicates that formal recognition is warranted.

In recent years there have been a number of taxonomic revisions and descriptions of new species of *Panulirus*. Berry (1974) broke down the widely distributed Indo-West Pacific *Panulirus homarus* (Linnaeus) into three geographic forms (which were given the rank of subspecies) distinguished by color and morphology. Another Indo-West Pacific species *Panulirus longipes* (A. Milne-Edwards), was found to be a complex of related species and subspecies by several authors: *P. pascuensis* Reed, *P. marginatus* (Quoy and Gaimard), *P. cygnus* George, *P. japonicus* (von Siebold), *P. longipes longipes* (A. Milne-Edwards), and *P. longipes femoristriga* (Von Martens) (Reed, 1954; George, 1962; George and Holthuis, 1965). The taxonomic status of *P. argus* has remained unchanged since it was originally described by Latreille in 1804, but recent observations have raised questions about the taxonomic status of this species.

Restriction fragment length polymorphism (RFLP) analyses of mtDNA (Silberman *et al.*, 1994a, b) revealed no evidence of genetic differentiation among populations of *Panulirus argus* from Venezuela to Bermuda, but high levels of mtDNA diversity were observed. In particular, Silberman *et al.* (1994a) observed three individuals (two from Miami and one from Venezuela) that displayed unusually high levels of mtDNA sequence divergence. The suspicion that *P. argus* in Brazil might

be biologically different from *P. argus* in the Caribbean and Western Atlantic (A. B. Williams, D. K. Camp, personal communications) led us to suspect that the three genetically differentiated lobsters observed by Silberman *et al.* (1994a) might have originated from genetically divergent Brazilian stocks. Therefore, we initiated a study that expanded previous work to include Brazilian populations of *P. argus*. The present study compares DNA sequence from portions of two mitochondrial genes in samples of *P. argus* from Brazil and samples of *P. argus* from other locations in the Caribbean. To gauge the taxonomic significance of the genetic differences observed between Caribbean and Brazilian *P. argus*, we included members of the closely related species of the japonicus-group of *Panulirus* (*P. longipes longipes*, *P. marginatus*, and *P. cygnus*) in the phylogenetic analyses.

MATERIALS AND METHODS

Panulirus argus mtDNA was obtained from the samples examined by Silberman *et al.* (1994a). A single sample from 5 representative locations in the Caribbean were used for DNA sequence analysis. Samples from Brazil were collected, but not included in Silberman *et al.* (1994a) because of technical problems with RFLP analysis. Samples of Brazilian *P. argus* were collected at 5°13'S, 35°10'W off Rio Grande do Norte region of Brazil. Tissue samples from Brazil were frozen prior to DNA isolation and 5 samples were used for sequence analysis. Collection locations are listed in Fig. 3. DNA

was also isolated from ethanol-preserved tissue samples of *P. longipes longipes* (from the Philippines), *P. marginatus* (Hawaii), and *P. cygnus* (Australia). Standard phenol/chloroform extraction techniques were used for DNA isolation (Hillis *et al.*, 1990). Portions of 2 mitochondrial genes (16S rRNA and cytochrome oxidase I) were amplified by the polymerase chain reaction (PCR) using conserved primers (16Sar and 16Sbr for 16S rRNA and COIa and COIf for cytochrome oxidase I, as in Palumbi *et al.*, 1991). The purified double-stranded amplification products were used as templates for DNA sequencing with both the L- and H-strand primers using the Δ Taq cycle sequencing kit (U.S. Biochemical Corporation, Cleveland, Ohio). Cycle-sequencing reactions were done using γ - 32 P-dATP end-labeled primers and approximately 0.1 μ g of DNA template. Sequences were determined in both directions for all sequences.

DNA sequences were aligned manually using the Seq-App DNA sequence editor and analysis program (D. G. Gilbert, Department of Biology, University of Indiana, Bloomington, Indiana 47405) (Figs. 1, 2). Phylogenetic analyses of aligned DNA sequences were done using maximum parsimony, maximum-likelihood, and neighbor-joining. For parsimony analyses, the branch and bound search algorithm in PAUP 3.1.1 (Swofford, 1993) with MULPARS and furthest addition settings were used. Branch and bound bootstrap analyses (250 replicates) were performed to evaluate the degree of support for particular nodes (Felsenstein, 1985). Maximum-likelihood analysis was done using the program DNAML in PHYLIP 3.57c (Felsenstein, 1995) with a transition/transversion ratio of 2.0. Neighbor-joining trees were constructed using MEGA (Kumar *et al.*, 1993), with Tamura-Nei distances and the pair-wise deletion option in effect. Estimates of nucleotide sequence divergence were calculated by the method of Jukes and Cantor (1969). *Scyllarides nodifer* (Stimpson) (Decapoda, Scyllaridae) and *Homarus americanus* (H. Milne Edwards) (Decapoda, Nephropidae) were used as outgroup taxa for phylogenetic analysis. For a review of the different approaches to phylogenetic analysis, see Swofford *et al.* (1996).

RESULTS

The (491) base pairs of the 16S gene contained (125) informative sites. The mean base content of the light strand of the 16S sequence was 30.2% A, 15.4% C, 22.3% G, and 32.1% T, indicating a moderate AT bias. The pattern of nucleotide substitution (excluding the outgroup) favored transitions (76) over transversions (55), yielding a transition/transversion (s/v) ratio of (1.3) and a range of (0.47)–(9.67). Nucleotide sequence divergence estimates from 6.3% (between *P. marginatus* and *P. cygnus*) to 22.3% (between Brazilian *P. argus* and *P. cygnus*). Mean nucleotide sequence divergence between Caribbean and Brazilian *P. argus* was 8.3%. Of (597)

base pairs determined for the COI gene, the 114 informative sites were evenly distributed over the length of the sequence (Fig. 2). There was also evidence of base composition bias for the COI sequence (26.6% A, 22.5% C, 18.1% G, 32.8% T). The number of informative sites were greater at both first and third codon positions than at second codon positions. The pattern of nucleotide substitution was biased in favor of transitions (121) over transversions (50), resulting in an s/v ratio of (2.42), with a range of (1.58)–(2.58). Estimates of nucleotide sequence divergence for COI ranged from 11.8–24.9%. Nucleotide sequence divergence between Caribbean and Brazilian *P. argus* was estimated to be 19.8%.

Phylogenetic analysis of 16S sequences generated four equally parsimonious trees of 396 steps (CI = 0.813, RI = 0.731). Three distinct clades were revealed in all four parsimony trees, with the nodes strongly supported by bootstrap values (>90%). Maximum-likelihood selected a tree that was identical to one of the four minimum-length parsimony trees (Fig. 3). All branch lengths were significant except two terminal branches (to *P. argus* Mi and *P. argus* BR1). The neighbor-joining tree differed from the maximum-likelihood tree only in the position of *P. argus* P and *P. argus* V, but was also the same as one of the trees selected by parsimony (results not shown). The three groupings supported by all phylogenetic analyses were Caribbean populations of *P. argus*, Brazilian *P. argus*, and the three species in the japonicus group of *Panulirus* (*P. longipes longipes*, *P. marginatus*, and *P. cygnus*). The only differences among the minimal-length trees was in the branching order among *P. argus* P, *P. argus* V, *P. argus* Mi, and *P. argus* J in the Caribbean clade of *P. argus*. This is not entirely surprising, since prior research has indicated that there is no evidence for genetic differentiation among Caribbean populations of *P. argus* (Silberman *et al.*, 1994a).

DISCUSSION

In recent years, many common and economically important taxa have been shown to be complexes of previously unrecognized species (Knowlton, 1993, and references

Fig. 1. Aligned nucleotide sequences for a region of the 16S rRNA gene. Sequences were aligned against *Homarus americanus* (Genbank accession number: U11238). Dots indicate identity and dashes indicate gaps.

<i>H. americanus</i>	??????????	?????ACCT	GCCCACTGGG	ATC--AAA--	CTAAAGGGCC	GCGGTATTTT
<i>S. nodifer</i>	TTTAATATCA	.GTCTGG...	T.-.A	TAT.....GT	T....-....	AA.
<i>P. argus</i> Mi	TTTTATGTA	AGTCTAG...	G.-.A	TAATTG...T	T...CC....	GCA.
<i>P. argus</i> P	TTTTATGTA	AGTCTAG...	G.-.A	TAATTG...T	T...C-....	GCA.
<i>P. argus</i> V	TTTTATGTA	AGTCTAG...	G.-.A	TAATTG...T	T...C-....	GCA.
<i>P. argus</i> A	TTTTATGTA	AGTCTAG...	G.-.A	TAATTG...T	T...C-....	GCA.
<i>P. argus</i> J	TTTTATGTA	GGTCTAG...	G.-.A	TAATTG...T	T...C-....	GCA.
<i>P. argus</i> BR1	TTTTATGTA	AGTCTA-...	G.T.A	TAATT..G.T	T...C-...-	GCA.
<i>P. argus</i> BR2	TTTTATGTTT	AGTCTAGG...	G.-.A	TAATT.TG.T	T...C-....	GCA.
<i>P. argus</i> BR3	TTTTATGTAC	AGTCTAG...	G.-.A	TAATT..G.T	T...C-..A.	GCA.
<i>P. argus</i> BR4	TTTTATGTAC	AGTCTAG...	G.-.A	TAATT..G.T	T...C-....	GCA.
<i>P. argus</i> BR5	TTTTATGTAC	AGTCTAG...	G.-.A	TAATT..G.T	T...C-....	GCA.
<i>P. longipes</i>	??????????	??????????	????G.-.A	TAT.TG..GT	T..GC-....	GCA.
<i>P. cygnus</i>	??TATGTAA	CGTCTAGA..	G.-.A	TAT.T...GT	TC..C-....	GCA.
<i>P. marginatus</i>	TTTTATGTAC	ACTCTAGA..	G.-.A	TAT.T...GT	TC..C-....	GCA.

<i>H. americanus</i>	AACCGTGCGA	AGGTAGCATA	GTCATTAGTC	TCTTAATTGG	AGGC-TTGTA	T-AATGGTTG
<i>S. nodifer</i>	G.....T.	A.A..T...	T....A...	G..G....	G..G....	G..G....
<i>P. argus</i> Mi	G.....T.	A....T...	CT.....A	G.A.C.G...	A..C..C..	A..C..C..
<i>P. argus</i> P	G.....T.	A....T...	CT.....A	G.A.-.G...	A..C..C..	A..C..C..
<i>P. argus</i> V	G.....T.	A....T...	CT.....A	G.A.-.G...	A..C..C..	A..C..C..
<i>P. argus</i> A	G.....T.	A....T...	CT.....A	G.A.-.G.A.	A..C..C..	A..C..C..
<i>P. argus</i> J	G.....T.	A....T...	CT.....A	G.A.-.G...	A..C..C..	A..C..C..
<i>P. argus</i> BR1	G.....T.	A....T...	C.....	G.A.-.G...	A..C..C..	A..C..C..
<i>P. argus</i> BR2	G.....T.	A....T...	C.....	C.A.-.G...	A..C..C..	A..C..C..
<i>P. argus</i> BR3	G.....T.	A....T...	C.....	G.A.-.G...	A..C..C..	A..C..C..
<i>P. argus</i> BR4	G.....T.	A....T...	C.....	G.A.-.G...	A..C..C..	A..C..C..
<i>P. argus</i> BR5	G.....T.	A....T...	C.....	G.A.-.G...	A..C..C..	A..C..C..
<i>P. longipes</i>	G.....T.	A....T...	C.....	G.A.-.G...	G..C..C..	G..C..C..
<i>P. cygnus</i>	G...C..T.	A....T...	C.....	G.A.-.G...	G..C..C..	G..C..C..
<i>P. marginatus</i>	G.....T.	A....T...	C.....	G.A.-.G...	G..C..C..	G..C..C..

<i>H. americanus</i>	GACAAGAAGC	AAACTGTCTC	-AAACACAAA	AATTGAATTT	GACTTTTAAG	TGAAAAGGCT
<i>S. nodifer</i>	A...A.G..	T.....	-T..G.T...	TC.....A..	A..A.....	
<i>P. argus</i> Mi	C.....AT	TGT.....	--CT.T...	T....A..	C..A.....	
<i>P. argus</i> P	C.....AT	TGT.....	-T.CT.T...	T....A..	C..A.....	
<i>P. argus</i> V	C.....AT	TGT...G...	-T.CT.T...	T....A..	C..A...T.	
<i>P. argus</i> A	C.....AT	TGT.....	-T.CT.T...	T....A..	C..A.....	
<i>P. argus</i> J	C.....AT	TGT.....	-T.CT.T...	T....A..	C..A.....	
<i>P. argus</i> BR1	C.....AT	TGT.....	CT.CT.T...	T....G..	C..A.....	
<i>P. argus</i> BR2	C.....TAT	TGT.....	-T.CT.T...	T...CG.C	C..A...C.	A.
<i>P. argus</i> BR3	C.....AT	TGT.....	-T.CT.T...	T....G..	C..A.....	
<i>P. argus</i> BR4	C.....AT	TGT.....	-T.CT.T...	T...G..C	C..A.....	
<i>P. argus</i> BR5	C.....AT	TGT.....	-T.CT.T...	T...G..C	C..A.....	
<i>P. longipes</i>	A.....AT	TG.....	-TGCT.T...	TT....A..	A..G.....	
<i>P. cygnus</i>	A.....AT	CG.....	-TGCT.T...	TC....A..	A..G.C....	
<i>P. marginatus</i>	A.....AT	CG.....	-TGCT.T...	TT.....	A..G.....	

<i>H. americanus</i>	TAAATATTTT	AAAGGGACGA	TAAGACCCTA	TAAAGCTTAA	TAATTTAGTA	TATAATTAG-
<i>S. nodifer</i>	GAGC.	.G.....		...T...T.	...T...GCCT	GGGT-
<i>P. argus</i> Mi	GGAA.	.GG.....		...T...T.	C...AG.GC	.T...G...TT
<i>P. argus</i> P	GGAA.	.GG.....		...T...T.	C...AG.GC	.T...G...TT
<i>P. argus</i> V	GGAA.	.GG.....		...T...T.	C...AG.GC	.T...G...TT
<i>P. argus</i> A	GGAA.	.GG.....		...T...T.	C...AG.GC	.T...G...TT
<i>P. argus</i> J	GAAA.	.GG.....		...T...T.	C...AG.GC	.T...G...TT
<i>P. argus</i> BR1	AAA.	.GG.....		...T...T.	C...GG.CC	.C.GGC.GTT
<i>P. argus</i> BR2	AAA.	.GG.....		...T...T.	C...GG.CC	.C.GGC.GTT
<i>P. argus</i> BR3	AAA.	.GG.....		...T...T.	C...GG.CC	.C.GGC.GTT
<i>P. argus</i> BR4	AAA.	.GG.....		...T...T.	C...GG.CC	.C.GGC.GTT
<i>P. argus</i> BR5	AAA.	.GG.....		...T...CT.	C...GG.CC	.C.GGC.GTT
<i>P. longipes</i>	GAGA.	.GG.....		...T...G.	.GG.CA.AAT	AT....TTT
<i>P. cygnus</i>	GAGA.	.GG.....		...T...G.	.G..CGG.GT	AGC....TTC
<i>P. marginatus</i>	AGA.	.GG.....		...T...G.	A.AGT	A...G...TTC

<i>H. americanus</i>	ATGAGTTGAA	AGTTTAATAT	TATTTATATA	CTA-?AATTA	TTTCGTTGGG	GCGACGATGA
<i>S. nodifer</i>	.AA.A..AGG	TTC.C.T.TA	AT.....AG	G.T-A...A-	...T.....	.A...AGA.G
<i>P. argusMi</i>	.GA.AG.AGT	TA.GG.TCTA	A..C.--AG	G.TTA...G	...T.....	AGAAG
<i>P. argusP</i>	.GA.AG.AGT	TA.GG.TCTA	A..C.--AG	G.TTAT...G	...T.....C	AGAAG
<i>P. argusV</i>	.GA.AG.AGT	TA.GG.TCTA	A..C.--GG	G.TTA...G	...T.....	AGAAG
<i>P. argusA</i>	.GA.AG.AGT	TA.GG.TCTA	A..C.--AG	G.TTG...G	...T.....	AGAAG
<i>P. argusJ</i>	.GA.AG.AGT	TA.GG.TCTA	A..C.--AG	G.TTA...G	...T.....	AGAAG
<i>P. argusBR1</i>	.GA.AC.AGT	TA.CG.TCTA	A..C.--GT	G.TTC...G	...T.....	AGAAG
<i>P. argusBR2</i>	.GA.AC.AGT	TA.CG.TCTA	A..C.--GT	G.TTC...G	C..G.....C	AGAAG
<i>P. argusBR3</i>	.GA.AC.AGT	TA.CG.TCTA	A..C.--GT	G.TTC...G	...T.....	AGAAG
<i>P. argusBR4</i>	.GA.AC.AGT	TA.CG.TCTA	A..C.--GT	G.TTC...G	...T.....	AGAAG
<i>P. argusBR5</i>	.GA.AC.AGT	TA.CG.TCTA	A..C.--GT	G.TTC...G	...T.....	AGAAG
<i>P. longipes</i>	...TAG.AGT	TA.AAGC.TA	A..C.--...T	T.TGG.C.AT	...T.....	AGG.G
<i>P. cygnus</i>	...TAG.AGT	TA.AA.C.TA	A..C.--...CTTG.T.A?	...T.....	AGG.G	
<i>P. marginatus</i>	.C.TAG.AGT	TA.AAGC.TA	A..C.--...G	..CTG.T.A?	...T.....	A.G.G
<i>H. americanus</i>	TATAATTT??	GTAAGTGT	AAATTTAAAA	TACAGAGATA	TTT??????G	TGTGT---AA
<i>S. nodifer</i>	C....AAAA-	CC.	T..-AGTTT.	AGTTA.A.AT	G..TAA---	.TGT.---
<i>P. argusMi</i>	AA--	C..	TGG--TTC.	..TTA.TTGT	...TATTAT.	GT.AGTTT..
<i>P. argusP</i>	AA--	C..	TGG--TTC.	..TTA.TGGT	...TATTAT.	GT.AGTTT..
<i>P. argusV</i>	AA--	C..	TGG--TTC.	..TTA.TTGT	..ATATTAT.	GTCAGTTG..
<i>P. argusA</i>	AA--	C..	TGG--TTC.	..TTA.TTGT	.G.TATTAT.	GT.AATTT..
<i>P. argusJ</i>	AA--	C..	TGG--TTC.	..TTA.TTGT	...TATTAT.	GC.AATTT..
<i>P. argusBR1</i>	AA--	C..	TGGC..TTT.	..TTA.TTGT	...TATTAT.	GTC-ATTTG.
<i>P. argusBR2</i>	AA--	C..	CGGC..TTT.	..TTA.TGGT	...TATTAT.	GTC-ATTTG.
<i>P. argusBR3</i>	AA--	C..	TGGC..TT-	..TTA.T?GT	...TATTAT.	GCC-ATTTG.
<i>P. argusBR4</i>	AA--	C..	CGGC..TT-	..TTA.TTGC	...TATTAT.	GTC-ACCTG.
<i>P. argusBR5</i>	AA--	C..	CGG...TT-	..TTA.TGGC	...TATTAT.	GCC-ATTTGG
<i>P. longipes</i>	A--	C..	TG..G...G	A.TTA.TTAT	...TATTGT.	.TGTGTTTG.
<i>P. cygnus</i>	A--	CC.	TGG-A...C	AG.TA.TTAT	...TATTGT.	.TATGTCT..
<i>P. marginatus</i>	AA-	CC.	TG..AG.G.T	AG.TA.TTAT	...TATTAT.	.TATGTTT..
<i>H. americanus</i>	?TGATCCTTG	?TTGT-TGAT	TAAATTTAA	GTTACTTTAG	GGATAACAGC	GTTATTTATT
<i>S. nodifer</i>	G.....T	AA.AA-G..A	ATT.GAA...			..A...CC..
<i>P. argusMi</i>	AA.....CT	T...G-A...	C.C.GAC...			..A..C.T.C
<i>P. argusP</i>	AA.....CT	C...G-A...	C.C.GAC...			C.A..C.T.C
<i>P. argusV</i>	AA.....CT	T...G-A...	G.C.GAC...			..C..C.C.C
<i>P. argusA</i>	AA.....CT	T...G-A...	C.C.GAC...			..A..C.T.C
<i>P. argusJ</i>	AA.....CT	T...G-A...	C.C.GAC...			..A..C.T.C
<i>P. argusBR1</i>	AA.....CT	T...G-A...	GTC.GAC...			..A..C.T.C
<i>P. argusBR2</i>	AA.....CT	C...GGA...	GTC.GAC...			..A.CC.T.C
<i>P. argusBR3</i>	AA.....CT	T...GGA...	GTC.GAC...			..C..C.T.C
<i>P. argusBR4</i>	AA.....CT	TC..GGA...	GCC.GAC...			..A..C.T.C
<i>P. argusBR5</i>	AC.....CT	T...GGA...	GTC.GAC...			.GA..C.TC.
<i>P. longipes</i>	--.....C	T...A-A...	ATC.GA.C..			..A..C.TC.
<i>P. cygnus</i>	--.....C	T...A-A...	ATC.GA.C..			..A..C.TCC
<i>P. marginatus</i>	--.....C	T...A-A...	ATC.GA.C..			..A..C.TCC
<i>H. americanus</i>	TT-GAGAGTT	CATAT-CGAC	AAAAAGTTT	-GCGACCTCG	ATGTTGAATT	-AAAAATT?-
<i>S. nodifer</i>	..-.....	A	..GG..A..	-.....		-...G..CC-
<i>P. argusMi</i>	..-.....	A	.G..GG...	T.G?.....	?	-...G.?CCC
<i>P. argusP</i>	..-.....	...C.T...A	.G..GG...	T.....	...C.....	-...G.CCC-
<i>P. argusV</i>	..-.....	...C...C..	.G.GGG...	T.G.....	...G...	-...CG.?CC-
<i>P. argusA</i>	..-.....	...T...A	.G..GG...	-.....	?	C...G.ACC-
<i>P. argusJ</i>	..-.....	...C...A	.G..GG...	-.....	...G..G...	?...G.CCC-
<i>P. argusBR1</i>	..-.....	..C...A	.G..GG...	G.....	...G...	T...G.ACC-
<i>P. argusBR2</i>	..-.....	..C...A	.G..GG...	-.....	...G...	-...G.?ACC
<i>P. argusBR3</i>	..-.....	..C..T...A	.G.GGG...	-.....	...TC...	-...G.ACC-
<i>P. argusBR4</i>	..-.....	..C...A	.G..GG...	-G...CG...	...C..C..	A..CG.CC?-
<i>P. argusBR5</i>	C.C.....	..C...A	.G..GG...	-.....	...C.....	-...G.ACC-
<i>P. longipes</i>	..-.....	A	.GG.GG.G..	-.....		-...G.ACC-
<i>P. cygnus</i>	..-.....C	A	.GG.GG.G..	-.....		-...G.ACC-
<i>P. marginatus</i>	..-.....C	A	.GG.GG.G..	-.....		-...G.ACC-

Fig. 1. Continued.

<i>H. americanus</i>	?CCATGGCGT AGGA
<i>S. nodifer</i>	TTT.C..T.C ..A.
<i>P. argus</i> Mi	TT?....T.. ..CC
<i>P. argus</i> P	-?.....T.. ..CC
<i>P. argus</i> V	T?.....G.. ..CC
<i>P. argus</i> A	TTT....T.. ..CC
<i>P. argus</i> J	C?.....?.. ..CC
<i>P. argus</i> BR1	??T....T.A ..CC
<i>P. argus</i> BR2	CT?.....A ..C
<i>P. argus</i> BR3	-.....?..A C.CC
<i>P. argus</i> BR4	-?.....T.A ..CC
<i>P. argus</i> BR5	C.....T.A ..CC
<i>P. longipes</i>	TTTG...T.C ..C.
<i>P. cygnus</i>	TTTG...T.. ..C.
<i>P. marginatus</i>	TTTG...T.. ..C.

Fig. 1. Continued.

therein). Modern molecular or biochemical techniques have been responsible for the recognition of many of these systematic and taxonomic problems, because often these species are difficult to distinguish based on morphological criteria (McDonald *et al.*, 1991; Sarver and Loudenslager, 1991; Sarver and Foltz, 1993). Prior to the present study, the taxonomic status of *P. argus* has not been questioned, but previous studies have not included populations from Brazil (Menzies and Kerrigan, 1979; Menzies, 1980; Silberman *et al.*, 1994a). The degree of genetic differentiation between Brazilian and Caribbean populations of *P. argus* revealed in the present study is consistent with the level of differentiation seen between subspecies or species in other groups of organisms (Avise, 1994). Based on the same region of 16S used in this study, Machado *et al.* (1993) estimated nucleotide sequence divergence between two species of *Penaeus* to be 11.1%. Using sequence analysis of the mitochondrial 12S rRNA gene, Palumbi and Benzie (1991) estimated 11% divergence between two different species of *Penaeus*. Knowlton *et al.* (1993) found an average of 10.5% sequence divergence for the mitochondrial cytochrome oxidase I gene (COI) between seven pairs of sister species of snapping shrimp (*Alpheus*) from either side of the Isthmus of Panama. Mean divergence estimates between Caribbean and Brazilian *P. argus* were 8.32% for 16S and 19.82% for COI.

While there is no accepted level of sequence divergence associated with various levels of taxonomic rank, the magnitude of nucleotide sequence divergence can be an effective gauge of taxonomic standing (Avise,

1994). To assess the significance of the observed sequence divergence in Brazilian *P. argus*, we examined sequence variation among three recognized species in the japonicus-group (George and Holthuis, 1965). Mean nucleotide sequence divergence estimates among *P. longipes*, *P. cygnus*, and *P. marginatus* for 16S was 7.3% and 13.0% for COI. These values are comparable to the levels of nucleotide sequence divergence observed between Caribbean and Brazilian *P. argus*. Phylogenetic analyses and estimates of sequence divergence indicate that Brazilian *P. argus* is genetically distinct from populations in the Caribbean. However, it is not always appropriate to base taxonomic decisions on mitochondrial gene sequences alone, because discrepancies can exist between gene trees and true species trees (see Avise, 1994). If an mtDNA gene tree is concordant with other information, such as nuclear gene trees or morphological data, then there is convincing evidence that the mtDNA gene tree accurately reflects true evolutionary history (Avise, 1994).

Undescribed color differences (and possibly morphological differences) between Brazilian and Caribbean *P. argus* have been known by lobster researchers and seafood importers for several years (A. B. Williams, D. K. Camp, and C. Sanchez, personal communications). The principal differences appear in the basic color pattern of *P. argus* that are especially evident on the abdomen. The characteristic spots on the tail of *P. argus*, as well as the characteristic lateral white banding pattern, are reduced in the Brazilian form. The Brazilian form also has a dark band running across the tail fan that is not present in the Caribbean form. This description should be viewed cautiously until an appropriate morphological description can be done, but a color photograph illustrating the color differences between Caribbean/western Atlantic and Brazilian *P. argus* can be obtained upon request.

Color has not traditionally been used as a character for taxonomic classification, since preservation quickly destroys the original color of the animal. However, it is a rather reliable character for distinguishing species of decapods that otherwise differ little in morphology (Bruce, 1975, 1978). Genetic studies have validated the species status of morphologically similar color forms (Huber,

<i>P. stylirostris</i>	TTC	GGT	CAT	CCA	GAA	GTA	TAT	ATT	TTA	ATT	TTA	CCA	GCT	TTT	GGT	ATA	ATT
<i>P. argus</i> BR	..T	.T.	...	..C	..G	..G	...	...	...	...	...	...	..G	...	..G	...	..C
<i>P. argus</i>	.A.	...	..C	..C	..G	..C	...	...	C.C	..C	..G	..C	..A	...	..A	...	..C
<i>P. cygnus</i>	...	..A	..C	..C	...	..C	...	...	...	...	C.G	..C	..A	...	..C	...	...
<i>P. marginatus</i>	???	???	??C	..C	...	..T	..C	...	C..	..C	C..	...	..A	...	..C	...	...
<i>P. longipes</i>	???	???	?..	..C	...	..T	..C	...	...	..C	C..	..C	..A	...	A..	...	...
<i>P. stylirostris</i>	TCA	CAT	ATT	ATT	AGC	CAA	GAA	TCA	GGT	AAA	AAA	GAA	GCA	TTC	GGA	ACA	CTT
<i>P. argus</i> BR	..T	...	...	G..	GCT	...	...	..T	AA.	...	..G	A..	C.T	..T	...	G.G	T.A
<i>P. argus</i>	..C	..C	...	G..	GC.	...	...	AAC	...	..G	A.G	C.G	..T	..G	G..	..A	...
<i>P. cygnus</i>	..T	..C	...	G..	GCA	...	...	..C	AAC	...	...	A.G	C.T	...	..G	G.C	..G
<i>P. marginatus</i>	..C	...	...	G.C	GCA	..G	...	..G	AAC	..G	...	A.G	C.T	..T	..G	G.C	AA.
<i>P. longipes</i>	..T	...	...	G..	GCA	...	...	..G	AAC	..G	...	A.G	C.T	..T	..T	G.C	..A
<i>P. stylirostris</i>	GGT	ATA	ATC	TAT	GCT	ATA	TTA	GCC	ATT	GGT	GTC	TTA	GGA	TTT	GTA	GTA	TGA
<i>P. argus</i> BR	..A	...	...	..C	..C	..G	..G	..A	...	..A	A.T	C..	...	..G	...	..C	...
<i>P. argus</i>	..G	..G	...	..C	...	...	..G	..A	...	...	A..	...	..G	..C	...	...	...
<i>P. cygnus</i>	..G	...	..T	...	..C	..G	C.G	..T	...	...	A.T	C..	...	..C	..T	...	...
<i>P. marginatus</i>	..A	...	..?	...	..G	C.G	..T	...	...	A.T	C..	...	...	..T	..C	...	...
<i>P. longipes</i>	..G	..G	..T	...	..G	C..	..T	...	...	A.T	C..	...	...	..T	..C	..G	...
<i>P. stylirostris</i>	GCG	CAT	CAT	ATA	TTT	ACA	GTA	GGA	ATG	GAT	GTC	GAT	ACT	CGT	GCT	TAT	TTT
<i>P. argus</i> BR	..A	...	..C	..G	...	...	..G	...	..A	..C	..?	..C	...	..A	..A	..C	..C
<i>P. argus</i>	..T	..C	..C	..G	...	...	...	...	..?	..G	..C	..C	..A	..A	..A	..C	..C
<i>P. cygnus</i>	..C	..C	..C	..G	...	...	..T	..T	..A	..C	..T	...	..A	..A	..A	...	...
<i>P. marginatus</i>	..A	..C	..C	...	...	...	..C	...	..A	..C	..T	..C	..A	..A	..A	...	...
<i>P. longipes</i>	..A	..C	..C	..G	...	...	..T	..T	..A	..C	...	...	..A	..A	..A	..C	...
<i>P. stylirostris</i>	ACA	TCT	GCT	ACA	ATA	ATT	ATT	GCT	GTA	CCT	ACA	GGT	ATT	AAA	ATC	TTC	AGT
<i>P. argus</i> BR	...	..A	..A	..C	...	...	...	..C	...	...	..A	...	..C	G.G	...	..A	...
<i>P. argus</i>	...	..A	..G	...	...	...	...	..C	..A	...	..G	...	..G	G.G	...	..G	...
<i>P. cygnus</i>	...	...	...	..C	...	...	...	..C	..C	...	..C	..G	...	...	G.A	...	..G
<i>P. marginatus</i>	...	...	..A	..C	...	...	...	..C	..T	...	..C	A.G	...	...	G.A	...	..G
<i>P. longipes</i>	...	...	..A	..C	...	...	...	..C	..T	..C	..C	..A	...	...	G.A	...	..G
<i>P. stylirostris</i>	TGA	TTA	GGT	ACC	TTA	CAT	GGT	ACT	CAA	CTT	AAC	TAT	AGT	CCT	TCT	CTT	ATT
<i>P. argus</i> BR	...	...	..C	...	C.C	..C	..A	G..	..T	T.C	..CT	CTA	..CA	..A	..C	...	C.G
<i>P. argus</i>	...	C..	..G	..T	C..	..C	..A	G..A	..C	T..	TCG	CTG	..C.	...	...	..C	T.G
<i>P. cygnus</i>	..G	...	..A	...	C.T	..C	..A	G..	..T	T..	..C.	CT.	..C.	..C	...	..C	C.G
<i>P. marginatus</i>	..?	...	..A	..AT	C.C	..C	..A	G..	..T	T.C	..C.	CTC	..CA	..A	..C	...	T.A
<i>P. longipes</i>	...	...	...	...	C??	..C	..G	G..	..T	T.C	..C.	CT.	..CA	..C	..C	...	C.A
<i>P. stylirostris</i>	TGA	GCT	CTA	GGA	TTT	GTA	TTT	TTA	TTT	ACT	GTA	GGA	GGT	CTA	ACC	GGA	GTC
<i>P. argus</i> BR	...	..A	TC.	...	...	A.C	..C	...	..CC	..G	..CG	..T	..A	...	..T	..T	A.T
<i>P. argus</i>	..G	..A	T..	...	..C	A.T	..C	C..	...	..A	...	...	...	...	..A	..C	A.T
<i>P. cygnus</i>	..G	A.C	...	...	..C	A.C	...	..G	...	..A	..G	...	..C	...	..A	..T	A.T
<i>P. marginatus</i>	...	..C	...	...	..C	A.C	...	...	...	..A	...	..G	..C	T..	..A	..?	A.T
<i>P. longipes</i>	...	C.C	..T	...	..C	A.C	...	...	..C	..A	...	..T	..A	T..	..G	..T	A.T
<i>P. stylirostris</i>	GTC	CTA	GCT	AAC	TCT	TCA	ATT	GAT	ATT	ATT	CTA	CAT	GAT	ACA	TAC	TAC	GTA
<i>P. argus</i> BR	..A	..T	..C	...	..A	..T	..C	..C	..C	...	..T	..G.	..C	..T	...	..T	...
<i>P. argus</i>	..T	T..	..A	...	..A	..T	...	..C	..C	...	..T	..C	..C	..T	...	..T	..G
<i>P. cygnus</i>	..A	...	..C	...	..A	...	...	..C	...	...	..T	..C	...	...	...	...	...
<i>P. marginatus</i>	..A	...	..?	...	..A	...	...	...	...	...	..C	..C	...	...	C..	...	...
<i>P. longipes</i>	..A	...	..C	...	..A	...	...	...	...	...	..T	..CC	ACG	...	C..	...	...

Fig. 2. Aligned nucleotide sequences for a region of the cytochrome oxidase gene. Spiny lobster sequences were aligned against *Penaeus stylirostris* (see Palumbi and Benzie, 1991). Dots indicate identity.

<i>P. stylirostris</i>	GTA	GCA	CAT	TTC	CAT	TAC	GTC	CTA	TCA	ATG	GGA	GCT	GTA	TTT	GGT	ATT	TTC
<i>P. argus</i> BR	...	..T	...	...	..C	..T	..T	...	..T	..A	..T	..A	...	...	...	..C	..T
<i>P. argus</i>	...	...	..C	...	..C	...	...	...	..C	..A	...	..A	...	..G	..C	..C	...
<i>P. cygnus</i>	..T	..C	...	...	..C	...	..T	...	..T	...	..T	..C	..AG	..C	...	...	..T
<i>P. marginatus</i>	..T	..T	...	...	..C	..T	..T	...	..T	..A	..T	..A	...	...	...	...	..T
<i>P. longipes</i>	..T	..C	...	...	..C	...	..T	...	..T	..A	..T	..A	...	...	...	...	..T
<i>P. stylirostris</i>	GCA	GGT	ATT	GCT	CAT	TGA	TTC	CCC	TTA	TTT	ACT	GGA	CTT	ACA	TTA	AAT	CCT
<i>P. argus</i> BR	T..T	..A	...	...	...	...	...	...	...	..C	..A	...	A..A	T..C	C..	...	..C
<i>P. argus</i>	..C	..A	..C	..C	...	...	...	...	C..	..C	..A	...	A..A	T..	...	..C	...
<i>P. cygnus</i>	..C	..G	..C	..AC	...	...	...	...	C..	...	..C	..T	A..A	T..	C..	..C	...
<i>P. marginatus</i>	..C	..A	..C	..A	...	...	...	...	C..	...	...	...	A..A	T..G	C..T	...	...
<i>P. longipes</i>	..C	..A	...	..A	..C	..T	...	...	C..	...	..C	..?	A..G	T..	C..	..C	..C
<i>P. stylirostris</i>	AAA	TGA	TTA	AAA	ATT	CAT	TTC	CTA	GTT	ATA	TTC	ATT	GGA	GTA	AAC	ATC	ACA
<i>P. argus</i> BR	...	...	...	..G	...	..A	..T	TC	..CA	...	..T	G..	..T	..G	...	..CA	..T
<i>P. argus</i>	..G	...	...	...	...	..G	...	TC	..CC	...	...	...	..?	..G	...	..CT	..C
<i>P. cygnus</i>	...	...	...	...	...	..G	..T	AC	..CG	..G	..T	G..A	..T	...	...	..CT	..T
<i>P. marginatus</i>	...	...	...	..G	..C	..G	...	AC	..CA	...	..T	G..?	..T	...	..T	..CT	..T
<i>P. longipes</i>	...	...	...	..G	..C	..G	..T	AC	..CA	...	..T	...	..T?	...	...	TCT	..T
<i>P. stylirostris</i>	TTT	TTC	CCT	CAA	CAT	TTC	TTA	GGT	CTT	AAT	GGT	ATG					
<i>P. argus</i> BR	...	C..	...	...	...	...	C..T	...	..A	..G	...	..?					
<i>P. argus</i>	...	...	..C	..G	A..	...	C..	..C	..A	..GG	..G	..?					
<i>P. cygnus</i>	..C	..T	..C	..G	..C	..T	...	...	..G	..GG	..?	???					
<i>P. marginatus</i>	..C	???	???	???	???	???	???	???	???	???	???	???					
<i>P. longipes</i>	..C	???	???	???	???	???	???	???	???	???	???	???					

Fig. 2. Continued.

1985). Several closely related species of *Panulirus* are distinguished largely on the basis of color. For example, while some morphological differences exist between members of the japonicus-group of *Panulirus*, they are not obvious. The most conspicuous differences are the color patterns of the live animals (George and Holthuis, 1965). The three subspecies of *P. homarus* are also distinguished largely on the basis of the color of live animals (Berry, 1974). Since scientists rarely have access to living material from distant locations, it is easy to see why closely related species groups that differ primarily on the basis of color have often been synonymized.

While the present data suggest a northern and southern form of *P. argus*, the geographic location of the boundary between the two forms cannot be defined without more extensive sampling. It is likely that the approximately 1,100 km of coastline separating the Caribbean and Brazilian coral reefs serve as a significant barrier to gene flow for many reef-associated species, such as *P. argus*. Along the coastline of South America, there is a disjunct distribution of the shallow-water coral reef fauna as a result of the influence of a series of rivers, beginning with the

Orinoco River delta in Venezuela and extending to near Fortaleza, Brazil. Thus, the coral reefs of Brazil are separated from those of the Caribbean by a large area that is characterized by reduced salinities, high turbidity, and large deposits of sediment near the mouths of the Orinoco, Amazon, Tocantins, and Parnaiba Rivers. In this region of coastline, the coral reef fauna is replaced by species characteristic of soft bottoms. This region of reduced salinities, high turbidity, and soft muddy bottoms could represent a significant barrier to larval dispersal and postlarval settlement, resulting in isolation of Caribbean *P. argus* and Brazilian *P. argus*. This is supported by the demonstration that reduced salinity and sediment deposits lower survivorship of postlarvae and significantly influence overall distribution patterns of *P. argus* (Herrnkind *et al.*, 1988; Field and Butler, 1994). In addition, while a number of Caribbean species are associated with the Brazilian reefs, there is also a significant endemic component in each region (Briggs, 1974).

Modern DNA-based characters and phylogenetic analyses are being used increasingly in systematic and taxonomic studies (see

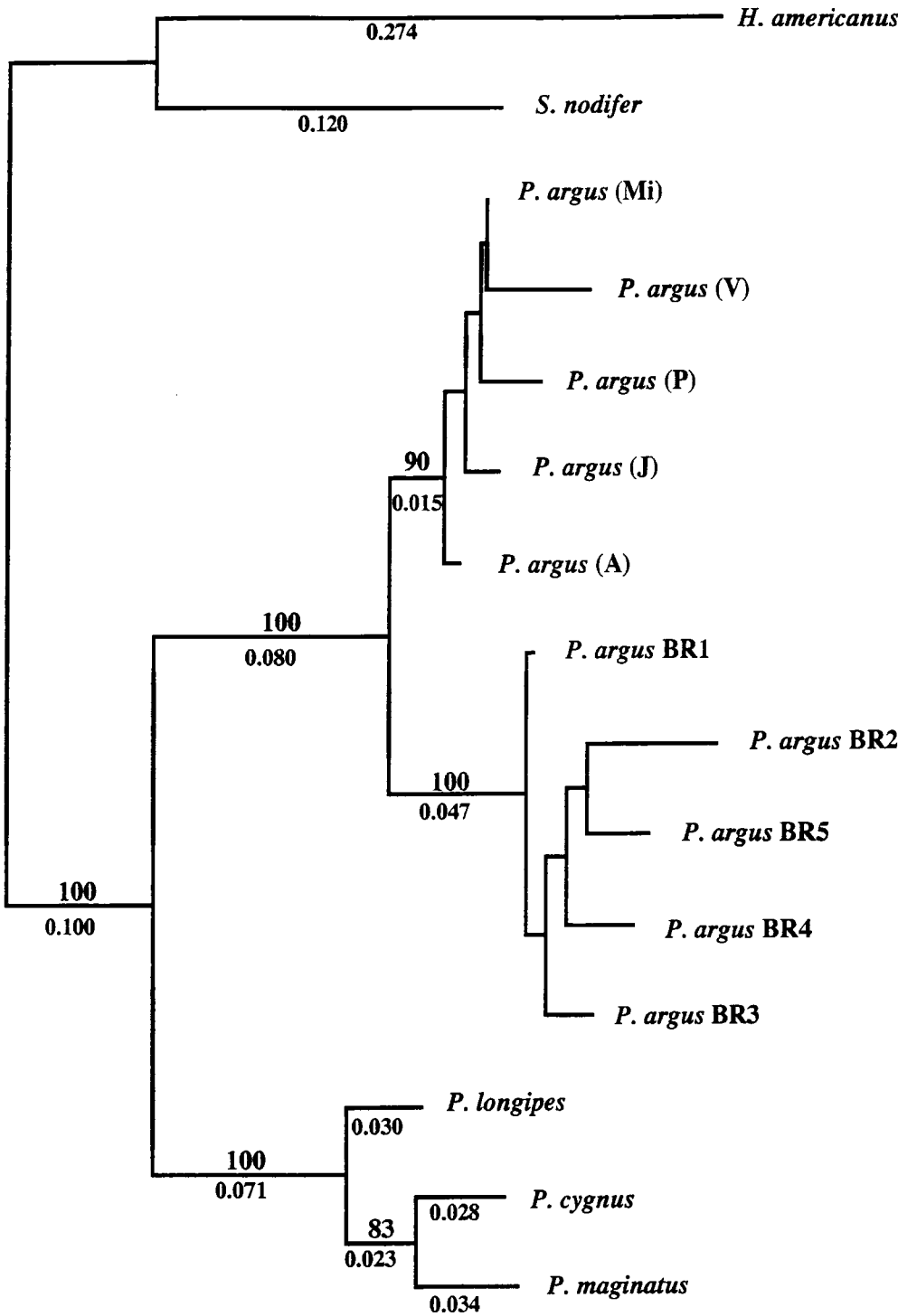


Fig. 3. Maximum-likelihood tree based on analysis of 16S rRNA gene sequences which is the same as one of the minimum-length trees resulting from maximum parsimony. Bootstrap values (250 replicates) for the parsimony tree are indicated over the internode branches and estimated branch lengths for the maximum-likelihood tree are indicated below the branches. Sample locations are as follows: *P. argus* A = Antigua, *P. argus* J = Jamaica, *P. argus* Mi = Miami, *P. argus* P = Panama, *P. argus* V = Venezuela, *P. argus* BR1–5 = Brazil.

Kornfield *et al.*, 1995). Molecular data are especially useful when large genetic differences and morphological differences are not concordant (Palumbi and Benzie, 1991). In cryptic or sibling species complexes, morphological differences can often be found once the genetically differentiated forms are recognized (e.g., Sarver and Foltz, 1993). The magnitude of mtDNA sequence differentiation observed between Brazilian and Caribbean/western Atlantic populations of *P. argus* indicate the existence of two genetically distinct groups diverged at the level seen between sister species. The mtDNA sequence data and the strong circumstantial evidence that there are corresponding color differences between Brazilian and Caribbean *P. argus* lead us to believe that the genetically distinct form of *P. argus* should be recognized. This is especially important, since identification of demographic and taxonomic units is essential for developing management guidelines for exploited species (Ryman and Utter, 1987; Collette and Vecchione, 1995). Since *P. argus* was originally described from the Caribbean and no previous scientific name is available for the Brazilian form (Holthuis, 1991), we suggest provisional recognition of two subspecies of *P. argus*, *P. argus argus* representing populations from Venezuela to Bermuda and *P. argus westonii*, new subspecies, representing populations of *P. argus* from Brazil.

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